

Humphrey Yang

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EDUCATION

Australian National University

Bachelor of Applied Data Analytics

GPA: 7/7; WAM: 88.7

Canberra, ACT

July 2019 – June 2023

Bachelor of Politics, Philosophy, and Economics

GPA: 6.75/7; WAM: 82.3

July 2018 – July 2021

Overall GPA: 6.9/7

Rewards: Chancellor's Letters of Commendation / Smarterknowledge Prize for Data Wrangling

Relevant Fields:

- **Data Analytics:** Statistical Learning | Mathematical Statistics | R, Python, and Java Programming | High Performance Computing | Population Analysis | Network Analysis | Data Wrangling and Mining

EMPLOYMENT HISTORY

Research Assistant | RSFAS, ANU

July 2023 – Now

- Assisting Professor Yanrong Yang in conducting research and experiments on the topic of **Benign Overfitting for Complicated Data Inference** under the ARC Discovery Project (DP230102250).
- Conducting large-scale simulations to investigate the model behaviour when applied to high-dimensional data using high-performance computing packages.

Research Assistant | QuantEcon

July 2022 – Now

- Contributed to the lectures and workshops on scientific computing and modelling in quantitative economics and specialised lectures using scientific and high-performance computing libraries (i.e., **Google JAX**) in Python.
- Contributed to projects such as QuantEcon.py library for quantitative economics and finance and application of Dynamic Mode Decomposition in income distribution.

Research Affiliate | Data 61, CSIRO

March 2023 – July 2023

- Conducted research to design attacks to produce responsible generative AI, especially in Large Language Models (LLMs) using **reinforcement learning and adversarial training**.
- Researched and designed reinforcement learning models for LLMs to induce undesirable behaviours from LLMs and employed appropriate metrics and framework to define action, state, and reward.

Summer Research Internship | Data 61, CSIRO

November 2022 – February 2023

- Researched to explore the landscape of **copyright protection in generative AI** and published a research poster to the **ACM Web Conference**.
- Implemented and measured **cutting-edge GAN and diffusion models** using CSIRO Bracewell HPC and improved their methodologies based on the results.

PUBLICATIONS AND RESEARCH PROJECTS

(* indicates equal contribution)

Published

Haonan Zhong*, Jiamin Chang*, **Ziyue Yang***, Tingmin Wu, Pathum Chamikara Mahawaga Arachchige, Chehara Pathmabandu, and Minhui Xue. 2023. Copyright Protection and Accountability of Generative AI: Attack, Watermarking and Attribution. In *Companion Proceedings of the ACM Web Conference 2023 (WWW '23 Companion)*, April 30-May 4, 2023, Austin, TX, USA

In Review

QuantEcon.py: A community-based Python library for quantitative economics ([link](#))
Journal of Open Source Software

In Work

Class Research Project

Economic Crisis and Migration Flow in Australia

Second Semester, 2021

- Delivered data visualisation on the **impact of major economic recessions since the 1930s on the inflow of permanent and temporary migration in Australia** based on the Australian Bureau of Statistics datasets on migration and indicators of labour market conditions.
- Evaluated migration intake reduction as a government policy during the economic recession by integrating datasets from multiple sources and performing **graphical data analysis and time series analysis (VAR model)**.

Insurance Redlining in Chicago on Racial Minorities 1977-1978

Second Semester, 2021

- Designed a generalised linear regression model to test the significance of race as a factor in insurance underwriting activities.
- Created visualisations of **GIS data using geojsonR and tmap packages** to supply evidence of insurance redlining on the racial minorities in Chicago.

Dissemination of Sentiment in Reddit Network

Second Semester, 2021

- Researched the **diffusion of sentiment** in subreddit r/Futurology and r/Collapse by mapping sentiment to the users' position in the network and analysed the variations in sentiment across different topics.
- Conducted sentiment analysis on the textual information and the sentiment to the social network using the quanteda package and analysed the key characteristics of the structure of the network using **Gephi and the igraph package**.

Epistemological Analysis of Propagation of Misinformation

First Semester, 2020

- Applied the **Infected-Susceptible-Recovered model simulation** in the **Epimodel** package to simulate the transmission of misinformation through different contagion methods.
- Employed linear regression and uncovered the infection rate (how contagious misinformation is) is insignificant in a group transmission, but the recovery rate (how hard it is to debunk misinformation) is significant in the further diffusion of misinformation.

TEACHING

RSFAS, Australian National University

Principles of Mathematical Statistics for Actuarial Studies (STAT6013) – Tutor

Semester 1, 2023

- Graduate course in mathematical statistics covering **Probability, Discrete and Continuous Probability Distributions, Multivariate Probability Distributions, Estimation, and Credibility Theory**.

Statistical Learning (STAT6040) – Tutor

Semester 1, 2023

- Graduate course in mathematical statistics covering research and developments in statistical concepts and high-performance statistical computing on topics including **Fundamental Theories in Statistical Learning and various parametric and non-parametric methods and model evaluation methods**.

Graphical Data Analysis (STAT3011/STAT4026/STAT7026) - Marking Assistant

Semester 2, 2022

- Graduate course in principles of data presentation and statistical graphics covering techniques for constructing **scientific visualisations**.

Quantitative Research Methods (STAT1008) – Tutor

Semester 2, 2022

- Undergraduate course in introductory mathematical statistics and statistical computing in R.

ACADEMIC REFEREES

Professor John Stachurski

Professor, Co-Chair of QuantEcon
Australian National University
Email: john.stachurski@anu.edu.au

Dr Yanrong Yang

Senior Lecturer, Research School of Statistics
Australian National University
Email: yanrong.yang@anu.edu.au

Dr Tina Wu

Research Scientist
CSIRO
Email: tina.wu@data61.csiro.au

ADDITIONAL INFORMATION

Technical skills:

- Proficient in Python, SQL, R, PyTorch, high-performance computing, Bash, LaTeX, and Markdown.
- Familiar with C, Java, Julia, and Sphinx.

Languages:

- English (Proficient; IELTS 8.5), Mandarin Chinese (Native).